

Explainable Multi-Disease Prediction using Machine Learning, SHAP, and LIME

Author

Megha K A
M.Sc. Data Analytics

1. Introduction

Artificial Intelligence and Machine Learning are increasingly being integrated into healthcare systems for early disease detection, clinical decision support, and predictive healthcare analytics. Although modern Machine Learning models achieve high predictive accuracy, many healthcare AI systems still suffer from a lack of interpretability.

In healthcare environments, clinicians require not only accurate predictions but also understandable explanations behind those predictions. Black-box models reduce trust and limit practical adoption in real-world medical systems.

This project presents a Multi-Disease Explainable AI framework for predicting:

- Diabetes Mellitus
- Chronic Kidney Disease (CKD)

using Machine Learning integrated with Explainable AI (XAI) techniques including:

- SHAP (SHapley Additive Explanations)
- LIME (Local Interpretable Model-Agnostic Explanations)

The project focuses on improving both predictive performance and model interpretability in healthcare AI systems.

2. Objectives

- The main objectives of this project are:
- Develop Machine Learning models for multi-disease prediction
- Predict Diabetes and Chronic Kidney Disease using healthcare datasets
- Compare predictive model performance
- Apply Explainable AI techniques for transparency
- Analyze clinically important healthcare features
- Improve trust and interpretability in healthcare AI systems

3. Datasets Used

3.1 Diabetes Dataset

Training Dataset:

- File Name: diabetes-train.csv
- Total Records: 70,692
- Features: 22
- Balanced Dataset

Testing Dataset:

- File Name: diabetes-test.csv
- Total Records: 253,680
- Features: 22
- Real-world imbalanced dataset

Important Features

- High Blood Pressure
- High Cholesterol
- BMI
- Smoking Habit
- Stroke History
- Physical Activity
- General Health
- Mental Health
- Age
- Income
- Education

Target Variable

Diabetes_binary

- 0 → No Diabetes
- 1 → Diabetes / Prediabetes

3.2 Chronic Kidney Disease Dataset

Dataset Information

- File Name: kidney_disease.csv
- Total Records: 400
- Clinical Features: 24
- Source: UCI Machine Learning Repository

Important Clinical Features

- Hemoglobin
- Specific Gravity
- Blood Urea
- Serum Creatinine
- Packed Cell Volume
- Hypertension
- Red Blood Cell Count
- Potassium

Target Variable

classification

- CKD
- Not CKD

4. Technologies Used

The following technologies and libraries were used:

- Python
- Pandas
- NumPy
- Scikit-learn
- XGBoost
- Matplotlib
- SHAP
- LIME
- Google Colab
- Jupyter Notebook

5. Methodology

5.1 Data Preprocessing

The preprocessing stage included:

- Cleaning categorical formatting inconsistencies
- Handling missing values
- Removing whitespace and tab anomalies
- Converting categorical variables into numerical form
- Mean imputation for missing data
- Train-test splitting

5.2 Machine Learning Models

The following Machine Learning models were implemented:

Diabetes Prediction

- Random Forest Classifier

CKD Prediction

- Decision Tree Classifier

Additional Ensemble Model

- XGBoost Classifier

6. Explainable AI Implementation

6.1 SHAP Explainability

SHAP analysis was used to interpret model predictions and identify globally important healthcare features.

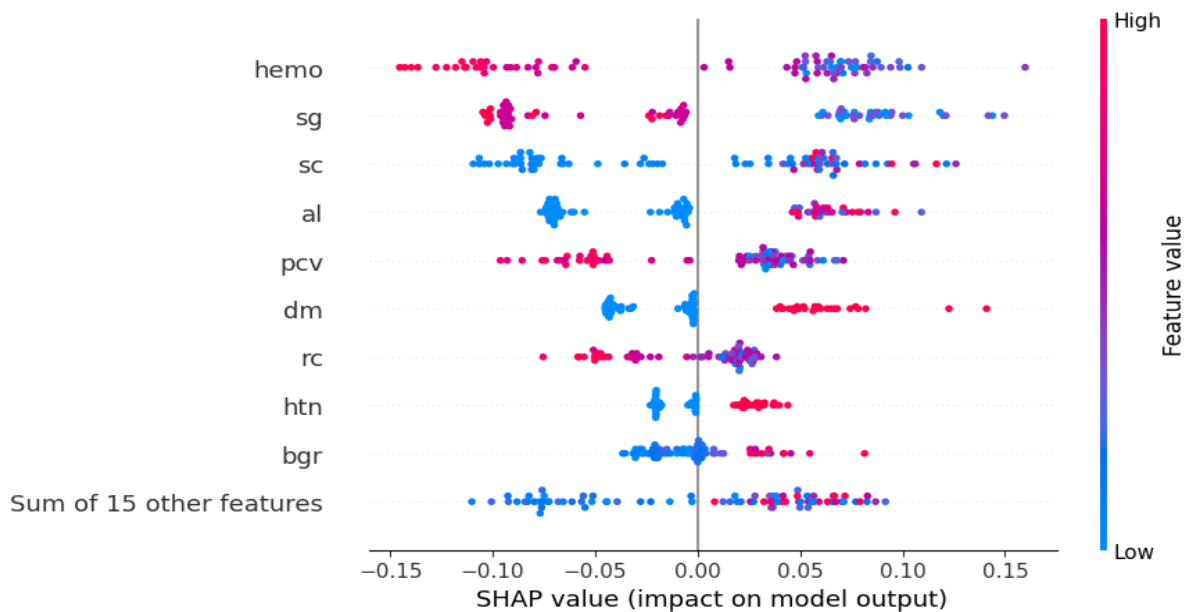


Fig 1: SHAP Summary Plot showing global feature importance influencing multidisease prediction

Important SHAP Features Identified

- Hemoglobin
- Specific Gravity
- Blood Urea
- Hypertension
- BMI
- General Health

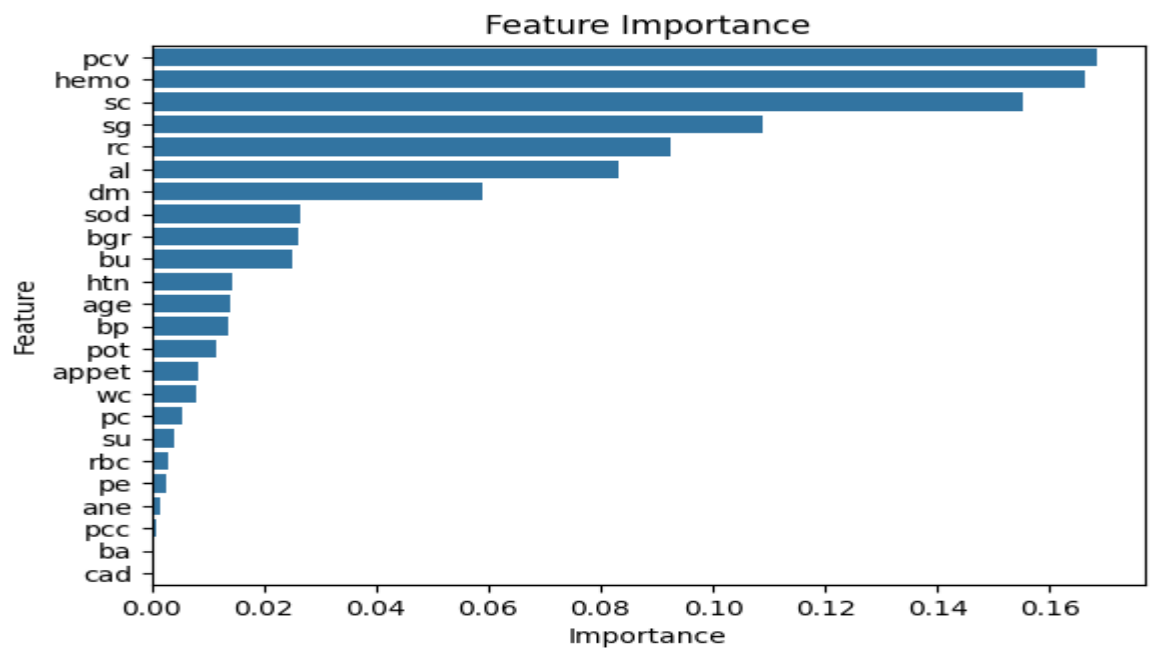


Fig 2: SHAP Feature Importance analysis identifying major healthcare attributes contributing to disease prediction

SHAP visualizations provided:

- Global feature importance
- Feature contribution analysis
- Transparent model interpretation

6.2 LIME Explainability

LIME was implemented to provide local prediction explanations for individual patient records.

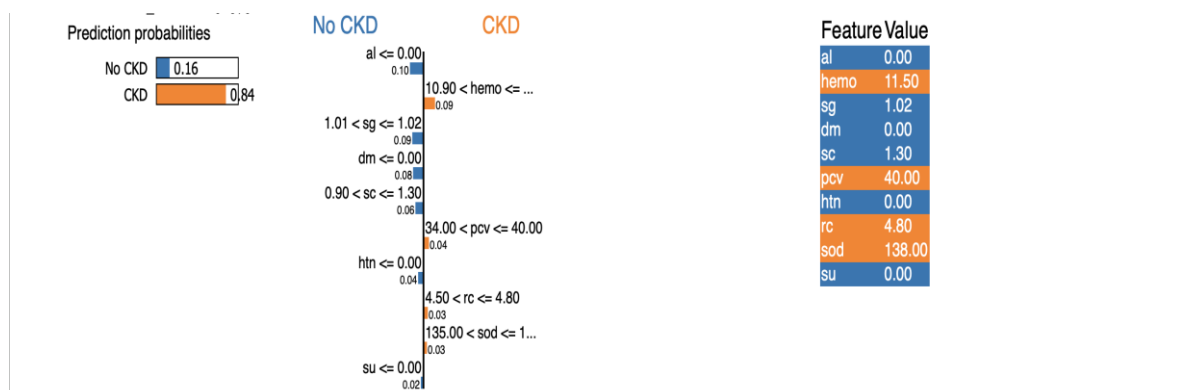


Fig 3: LIME explanation showing local feature contributions for an individual CKD prediction instance

The LIME analysis displayed:

- Prediction probabilities
- Local feature importance
- Patient-level interpretability

Important Features Identified by LIME

- Albumin
- Hemoglobin
- Specific Gravity
- Serum Creatinine
- Packed Cell Volume
- Hypertension

LIME improved transparency by explaining how specific healthcare features contributed to individual CKD predictions.

7. Model Performance

7.1 Diabetes Prediction Performance

Metric	Score
Accuracy	0.778

7.2 CKD Prediction Performance

Metric	Score
Accuracy	1.00
Precision	1.00
Recall	1.00
F1 Score	1.00
Cross Validation Score	0.97

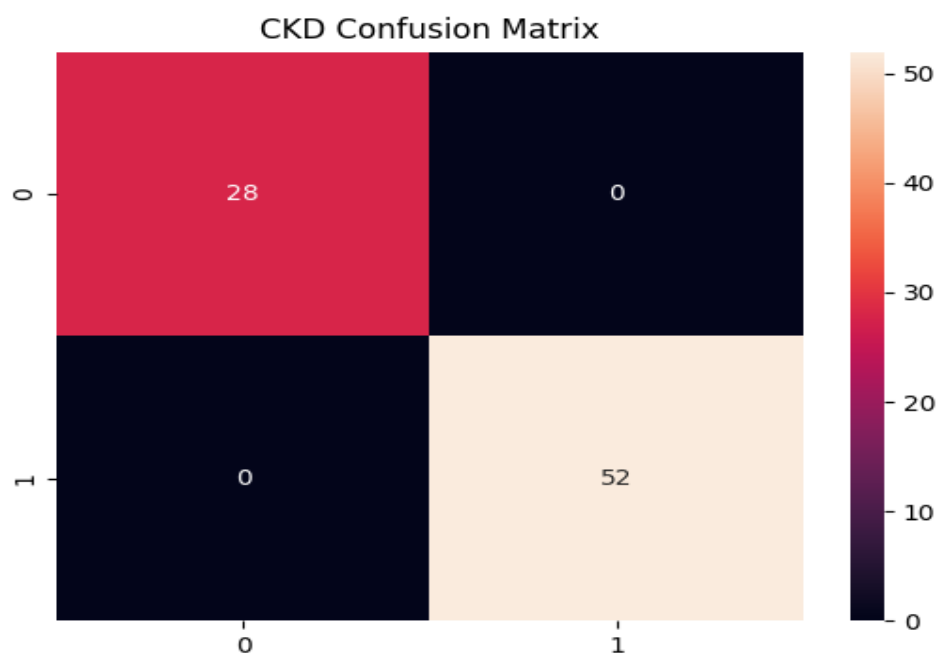


Fig 4: Confusion Matrix demonstrating classification performance of the healthcare prediction model

7.3 XGBoost Model Performance

Metric	Score
Accuracy	0.9875

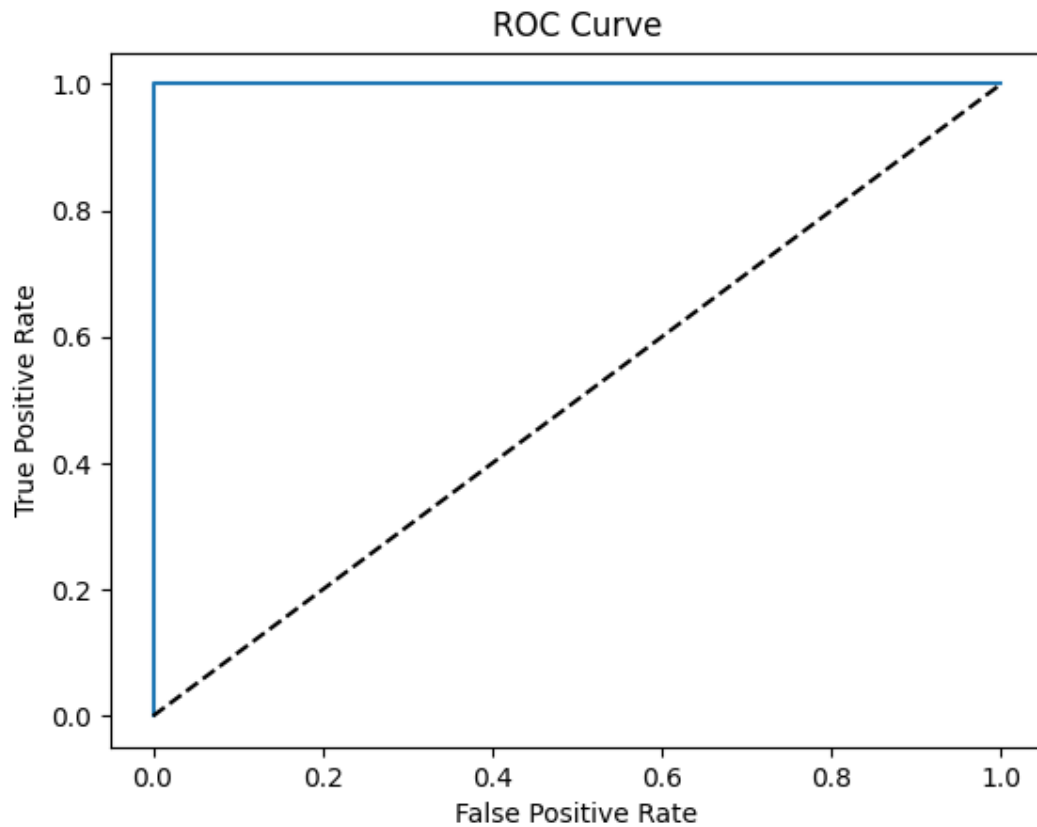


Fig 5:ROC Curve showing classification capability of the multidisease prediction framework

8. Visualizations Generated

The following visualizations were generated during implementation:

- SHAP Summary Plot
- SHAP Feature Importance Plot
- LIME Explanation Visualization
- ROC Curve
- Confusion Matrix
- Decision Tree Visualization
- Feature Importance Analysis

These visualizations improved interpretability and transparency of the healthcare prediction models.

9. Key Findings

The project demonstrated that:

- Explainable AI improves healthcare model transparency
- SHAP effectively identifies globally important clinical features
- LIME provides meaningful patient-level explanations
- Ensemble and Decision Tree models achieved strong predictive performance

- Interpretability is essential for trustworthy healthcare AI systems

10. Challenges Faced

Several implementation challenges were encountered:

- Missing values in CKD dataset
- Data formatting inconsistencies
- Class imbalance in diabetes dataset
- Clinical feature preprocessing complexity
- Explainability visualization integration

These challenges were resolved through preprocessing, imputation, and encoding techniques.

11. Future Improvements

Potential future enhancements include:

- Deep Learning healthcare models
- Real-time disease prediction systems
- Federated Learning for healthcare privacy
- Multi-class disease prediction
- Hyperparameter optimization
- Clinical deployment validation
- Explainable Deep Learning frameworks

12. Conclusion

This project successfully developed a Multi-Disease Explainable AI framework for predicting Diabetes and Chronic Kidney Disease using Machine Learning integrated with SHAP and LIME explainability techniques.

The implementation demonstrated that Explainable AI can significantly improve transparency, interpretability, and trust in healthcare AI systems while maintaining strong predictive performance.

The project highlights the importance of balancing predictive accuracy with interpretability in clinical AI applications and provides a strong foundation for future research in trustworthy healthcare AI systems.

13. References

[1] Lundberg, S.M., & Lee, S.I. "A Unified Approach to Interpreting Model Predictions," Advances in Neural Information Processing Systems, 2017.

[2] Ribeiro, M.T., Singh, S., & Guestrin, C. “Why Should I Trust You? Explaining the Predictions of Any Classifier,” ACM SIGKDD, 2016.

[3] Scikit-learn Documentation. Available: <https://scikit-learn.org>

[4] SHAP Documentation. Available: <https://shap.readthedocs.io>

[5] LIME Documentation. Available: <https://lime-ml.readthedocs.io>

[6] UCI Machine Learning Repository – Chronic Kidney Disease Dataset.